

KAUHAHA, Finland

WMO: 029130

Lat: 63.136N

Lon: 23.053E

Elev: 151

StdP: 14.62

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-14.8	-9.0	-19.2	1.9	-14.6	-13.0	2.7	-8.8	23.2	29.5	20.7	30.5	2.4	50	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.8	79.9	64.7	76.5	63.1	73.2	61.4	67.4	75.3	65.2	72.8	63.3	70.6	7.9	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	91.9	70.6	62.4	84.8	68.7	60.3	78.5	66.3	31.9	75.2	30.2	72.6	28.8	70.4	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.2	17.8	15.9	DB	-19.5	84.2	8.8	3.8	-25.8	87.0	-31.0	89.2	-35.9	91.4	-42.2	94.2	(4)
(5)			WB	-19.5	70.1	8.6	2.9	-25.7	72.2	-30.8	73.9	-35.6	75.6	-41.9	77.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	39.3	19.2	18.8	26.0	37.9	47.7	56.3	61.4	58.2	49.7	39.3	31.1	24.3
	DBStd	17.14	12.60	13.00	9.48	6.71	7.61	6.08	5.54	5.69	6.40	7.33	8.78	11.74
	HDD50	4875	954	874	743	367	135	12	1	5	84	336	567	797
	HDD65	9438	1419	1294	1208	814	538	269	139	222	460	796	1017	1262
	CDD50	960	0	0	0	3	64	202	355	258	74	4	0	0
	CDD65	49	0	0	0	0	2	8	29	10	0	0	0	0
	CDH74	572	0	0	0	0	39	101	313	119	0	0	0	0
	CDH80	91	0	0	0	0	3	17	60	11	0	0	0	0
Wind	WSAvg	7.1	7.3	7.2	7.5	7.4	7.9	7.3	6.1	6.1	6.6	7.1	7.5	7.6
Precipitation	PrecAvg	22.0	1.4	1.1	1.1	1.1	1.7	2.4	3.0	2.6	2.0	2.1	1.9	1.7
	PrecMax	27.2	2.3	1.9	1.9	2.8	3.1	4.7	4.8	4.8	5.4	4.6	3.5	2.9
	PrecMin	17.0	0.5	0.1	0.2	0.2	0.5	0.8	0.9	0.8	0.7	0.6	0.4	0.7
	PrecStd	2.6	0.5	0.6	0.4	0.6	0.8	1.0	1.1	1.1	1.3	1.0	0.8	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.1	42.1	49.7	65.4	78.7	82.5	85.8	81.1	70.4	57.6	50.1	44.8
		MCWB	38.5	38.2	41.6	50.4	59.3	64.7	68.3	65.9	60.2	53.5	48.4	44.1
	2%	DB	37.7	38.7	44.3	59.2	73.2	76.7	81.2	77.4	66.5	54.2	46.8	41.8
		MCWB	36.4	36.3	38.5	47.8	57.3	61.6	65.9	64.1	58.5	51.5	45.4	40.8
	5%	DB	35.9	36.0	40.8	54.1	67.9	72.7	77.3	73.6	63.2	52.0	44.7	39.5
		MCWB	34.9	34.8	36.3	44.8	54.7	59.2	64.4	62.6	56.5	49.8	43.7	38.1
	10%	DB	34.1	34.2	37.7	50.2	62.6	69.5	73.7	70.1	60.6	49.9	42.5	37.4
		MCWB	33.4	33.3	34.6	42.9	51.4	57.9	62.8	61.0	55.4	48.0	41.5	36.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.0	38.6	42.5	52.9	62.1	67.4	71.2	69.8	62.1	54.6	48.5	43.9
		MCDB	39.4	40.9	48.3	61.5	73.5	75.7	80.9	77.2	67.7	56.7	49.8	44.2
	2%	WB	36.6	36.5	39.1	49.1	59.1	64.2	68.5	66.3	59.8	52.3	45.6	41.0
		MCDB	37.3	37.9	43.0	57.4	69.9	73.8	77.0	73.0	64.6	54.0	46.5	41.4
	5%	WB	35.0	35.0	36.9	46.1	55.7	61.3	66.3	64.0	57.9	50.3	43.5	38.3
		MCDB	35.7	35.9	39.6	52.6	66.2	69.0	73.9	70.8	62.0	51.9	44.3	39.1
	10%	WB	33.5	33.4	35.0	43.1	52.5	59.2	64.3	62.0	56.2	48.1	41.3	36.3
		MCDB	34.2	34.0	37.2	49.4	61.2	66.9	71.8	68.5	59.9	49.5	42.0	37.1
Mean Daily Temperature Range	5% DB	MDBR	11.9	13.2	15.7	16.6	19.5	18.4	18.8	19.0	16.5	10.9	8.1	10.3
		MCDBR	9.1	10.2	16.4	24.8	28.8	25.2	25.3	24.9	20.5	12.3	8.9	8.2
		MCWBR	8.8	8.9	11.2	14.9	14.8	12.6	12.6	13.3	13.0	9.6	8.4	7.9
	5% WB	MCDBR	9.3	9.5	14.2	22.3	25.7	21.0	21.7	21.1	17.3	11.0	8.6	8.1
		MCWBR	9.2	8.4	10.3	14.2	14.0	11.8	11.7	12.3	11.6	9.3	8.3	8.0
Clear-Sky Solar Irradiance	taub	0.232	0.290	0.316	0.356	0.352	0.351	0.373	0.359	0.333	0.315	0.379	0.353	
	taud	2.041	2.169	2.253	2.333	2.431	2.456	2.416	2.451	2.509	2.385	2.688	2.593	
	Ebn at Noon	124	202	247	261	274	276	266	259	245	196	115	69	
	Edh at Noon	10	20	27	32	31	31	32	28	22	16	9	6	
All-Sky Solar Radiation	RadAvg	52	232	706	1181	1591	1736	1610	1233	720	279	69	24	
	RadStd	6	48	73	121	170	152	155	127	78	54	12	2	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.26	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		+0.81	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page